Inverse Trigonometric Function

Mind map

Meaning

- Restrict to a bijection
- $\sin^{-1} x$ = angle, not $\frac{1}{\sin x}$

Domain and range

- $\sin^{-1}$: $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$, $\cos^{-1}$: $[0, \pi]$
- $\tan^{-1}$: $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, $\cot^{-1}$: $(0, \pi)$
- $\sec^{-1}, \operatorname{cosec}^{-1}$: $|x| \geq 1$

Graph facts

- $\uparrow \sin^{-1}, \tan^{-1}$; $\downarrow \cos^{-1}, \cot^{-1}$
- Odd: first family
- Inflection at $x = 0$

Inequalities

- $\sin^{-1}, \tan^{-1}$: sign kept
- $\cos^{-1}, \cot^{-1}$: sign reversed
- Constant outside range: all or none

Evaluation

- Right triangle for θ
- $\sin(2\tan^{-1} x) = \frac{2x}{1+x^2}$

Property 1

- $\sin(\sin^{-1} x) = x$, $|x| \leq 1$
- $\tan(\tan^{-1} x) = x$, all x

Property 2

- $\sin^{-1}(\sin x)$: nearest $n\pi$
- $\cos^{-1}(\cos x) = |x - 2k\pi|$
- $\tan^{-1}(\tan x) = x - n\pi$
- Graphs: zig-zag, saw-tooth

Substitution

- $x = \sin\theta, \cos\theta, \tan\theta$
- $\sin^{-1} \frac{2x}{1+x^2} = 2\tan^{-1} x$, $|x| \leq 1$
- $\sin^{-1}(3x - 4x^3) = 3\sin^{-1} x$

Property 3 (negative x)

- $\sin^{-1}(-x) = -\sin^{-1} x$
- $\cos^{-1}(-x) = \pi - \cos^{-1} x$

Property 4

- $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$
- $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$
- Range of $(\sin^{-1} x)^2 + (\cos^{-1} x)^2$: put t

erty 5

- $x = \sin^{-1} \frac{1}{x}$
- $x = \pi + \tan^{-1} \frac{1}{x}$, $x < 0$

Addition / subtraction

- $\tan^{-1} \frac{x+y}{1-xy}$ ($+\pi$ if $xy > 1$)
- $\tan^{-1} \frac{x-y}{1+xy}$
- Factor $\frac{1}{2}$: half-angle

Equations

- One ITF, take trig of both sides
- Check roots in ranges
- Count roots by graphs

Series

- $T_r = \tan^{-1} b - \tan^{-1} a$
- Telescoping: first and last terms